

A Pilot Study of Smartphone Eye Tracking for Detection of Positional Nystagmus

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Keywords

Positional nystagmus · Smartphone eye tracking · EyePhone · Benign paroxysmal positional vertigo · Vestibular diagnostics

Abstract

Introduction: Detecting positional nystagmus is essential for diagnosing benign paroxysmal positional vertigo (BPPV). Therefore, developing methods to streamline this diagnosis can improve timely patient management and help prevent unnecessary emergency department visits. We aimed to evaluate the accuracy of a smartphone eye-tracking application in quantifying eye movements during positional testing to detect positional nystagmus. **Methods:** We recruited patients with positional dizziness suspected of having BPPV from the vestibular rehabilitation clinic and the consult service for dizzy patients (Tele-Dizzy) at Johns Hopkins Hospital. Using an in-house smartphone app (EyePhone), we recorded eye movements during the Dix-Hallpike and supine roll tests. Two expert clinicians reviewed the videos, and a third one adjudicated the

disagreements. Eye position data obtained from the EyePhone app were analyzed with an embedded algorithm to identify positional nystagmus. Using the adjudicated expert review as the reference standard, we evaluated EyePhone's accuracy in detecting positional nystagmus by calculating the sensitivity, specificity, and predictive values. **Results:** We recruited ten participants, 60% women, with an average age of 61.8 years. We reviewed 23 positional eye movement videos of participants while undergoing positional testing. The final adjudicated expert review identified positional nystagmus in 3 (13%) videos. The phone application traces indicated nystagmus in all 3 of these videos (sensitivity = 100% [95% CI = 44–100%]) and correctly ruled it out in 20 traces (specificity = 100% [95% CI = 84–100%]). The app demonstrated a positive predictive value of 100% (95% CI = 43–100%) and a negative predictive value of 100% (95% CI = 84–100%). **Conclusions:** This small pilot study shows proof-of-concept that a smartphone eye-tracking app without special phone attachments can detect positional nystagmus.

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Introduction

Benign paroxysmal positional vertigo (BPPV) is a leading cause of positional dizziness, with 2.4% prevalence in life time and commonly presenting as brief episodes of vertigo triggered by head movements [1]. The prevalence of this condition rises with age, peaking at the 61–70-year-old age group [2]. The diagnosis of BPPV relies on detecting positional nystagmus during positional maneuvers such as Dix-Hallpike and supine roll tests [3]. However, expertise is required for accurate test performance and recognition of subtle nystagmus and differentiating BPPV from other potentially more dangerous causes of dizziness [4]. In recent years, the introduction of quantified eye-tracking methods (e.g., video oculography [VOG]) has addressed some of these challenges and offers a more reliable method of evaluation and diagnosis of BPPV. Moreover, VOG can be used in the telemedical evaluation of patients with positional vertigo where experts are not readily available at the bedside [5]. Nonetheless, both the expertise and use of VOG are currently limited, and assessments indicate that a considerable number of patients with BPPV do not receive the standard evaluations outlined in the guidelines [6]. Furthermore, studies show that diagnostic challenges in emergency settings often lead to overuse of imaging – with inferior diagnostic yield, emergency department revisits, and delayed diagnoses [7, 8]. To address some of these challenges, researchers have studied using mini-VOG goggles that patients could use at home, which have shown promise in detecting positional nystagmus [9]. On the other hand, as smartphones have become ubiquitous in modern day life, there has been a growing interest in utilizing mobile phone technologies to diagnose conditions like BPPV in a scalable and cost-effective fashion [10]. Several studies have explored the use of smartphone video recording of the eyes for diagnosis of BPPV and other conditions with ictal dizziness [10, 11]. These studies show that smartphones can successfully capture videos of ictal and positional nystagmus [12–14]. Moreover, a study specifically looking into the accuracy of remotely reviewing smartphone-captured videos in detecting BPPV showed a sensitivity of 93% and a specificity of 100% [12].

Recent developments in smartphone hardware have made it possible to obtain eye position data using application programmer interfaces (APIs) such as Apple's augmented reality kit (ARKIT) [15]. Studies have shown the gaze estimation data provided by this tool kit to be

comparable to portable eye tracking [16]. Using the ARKIT API, our team has developed an in-house eye-tracking application called EyePhone [17]. We have previously published on the development of this application and how we calibrated eye and head tracking data based on standardized measurements [18]. Furthermore, we have previously shown that our smartphone app (EyePhone) can accurately characterize induced nystagmus in healthy volunteers in controlled research environments [19]. In this study, we sought to assess the accuracy of the EyePhone smartphone eye-tracking application in detecting positional nystagmus during standard positional maneuvers.

Methods

We prospectively recruited adult patients with a complaint of positional dizziness suspected of BPPV from the Johns Hopkins Vestibular Rehabilitation Clinic and our consult service for dizzy patients (Tele-Dizzy) who underwent positional testing as a part of their evaluation. We used the convenient sampling method (consecutive recruitment) for this pilot study. We excluded patients with a previously diagnosed ocular motor abnormality causing nystagmus, and those who could not undergo positional testing. As a part of the standard evaluation, the clinician performed positional testing to detect positional nystagmus. While patients were undergoing the positional testing, a study team member used the smartphone-based eye-tracking app (EyePhone) to record eye movements (Fig. 1). The application was installed on an Apple iPhone device (Model X or later) and utilized the front camera to obtain recordings with 1080p resolution and a frame rate of 60 frames per second. We utilized Apple's ARKIT to obtain eye position traces from video recordings of the face using the front camera. In our previous publications, we have extensively described how the app was developed [17], how head and eye tracking was performed and validated [20], and how we were able to utilize our in-house algorithm to detect and quantify induced nystagmus [19]. In this study, we used the same methodology for detecting nystagmus.

To compare the accuracy of the application in picking up characteristic eye movements, eye position traces were extracted from the app and analyzed using a MATLAB algorithm to detect positional nystagmus. This algorithm extracts the velocity of slow phases by detecting the quick phases using an iterative process, averaging the remaining samples, and then decides on the

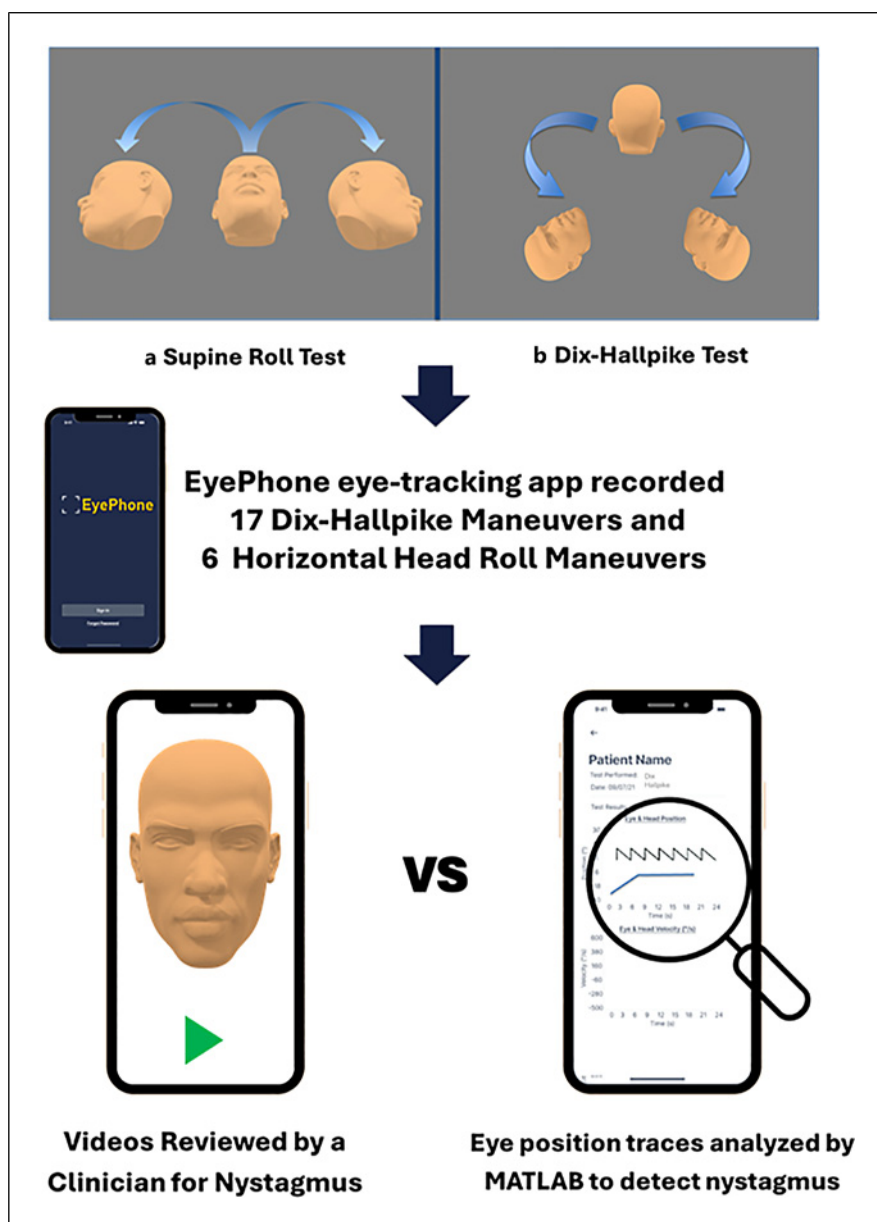


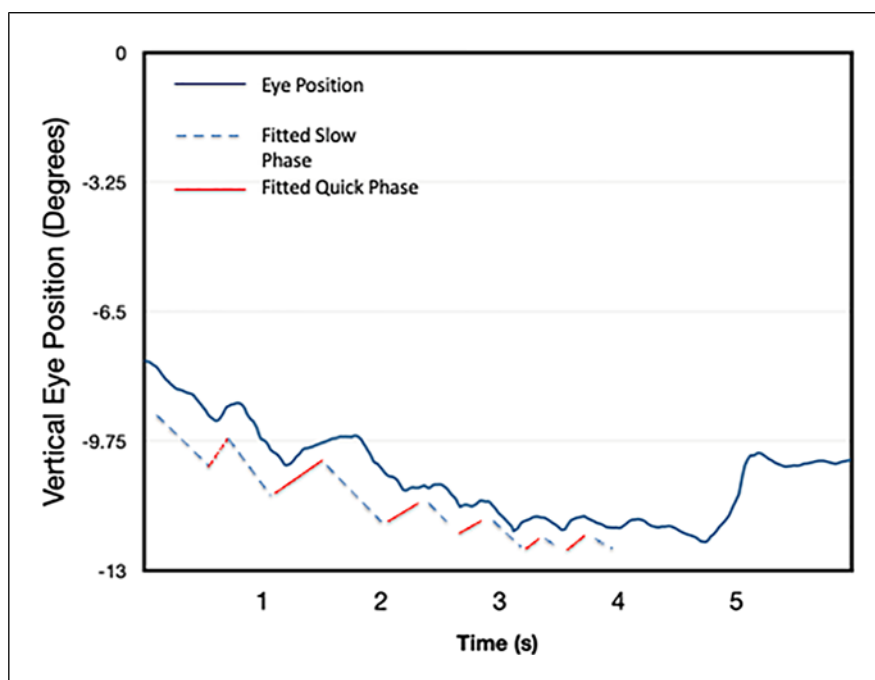
Fig. 1. Overview of our study procedure using the smartphone-based eye-tracking app (EyePhone) to record eye movements during the supine roll (a) and the Dix-Hallpike (b) tests.

presence/absence of nystagmus by applying a hard threshold to the resulting velocity [19]. Two independent expert reviewers masked to the outcome of the app's algorithm reviewed videos obtained by the app and labeled the ones with nystagmus. A third expert clinician was used to adjudicate the disagreements. The final adjudicated expert review was used as the reference standard, and we calculated the sensitivity, specificity, and positive and negative predictive value. Due to our small sample size, we used the Wilson method to calculate the 95% confidence intervals for each accuracy index [21].

Results

We recruited ten participants for the study, seven from the vestibular rehabilitation clinic and three from our Tele-Dizzy consult service (online suppl. Table 1; for all online suppl. material, see <https://doi.org/10.1159/000547008>). One participant could not undergo recording with the phone due to scheduling conflicts. The study team successfully obtained 23 recordings of positional tests (17 Dix-Hallpikes, 6 supine rolls) from the remaining nine participants. The average age of the participants included in the analysis was 59.6 ± 14.7

Fig. 2. A vertical eye position trace obtained using the EyePhone app reveals decaying upbeat nystagmus in a patient with clinically confirmed positional nystagmus consistent with posterior canal BPPV. The solid blue line represents the eye position as recorded by the application, while the dashed blue line indicates the slow phase of the nystagmus identified by the algorithm. The solid red lines denote the quick phases, as detected by the algorithm. BPPV, benign paroxysmal positional vertigo.



and 5 (55%) were females. None of the participants had any baseline neurological conditions when evaluated, most common medications used by the participants were blood pressure lowering (4 out of 9) and statins (4 out of 9). One of the participants reported taking meclizine (a vestibular suppressing medication for the dizziness).

Expert review of the videos identified nystagmus in 3 (13%) out of 23 videos. The traces obtained by the app correctly indicated the presence of nystagmus in all 3 of these traces (sensitivity = 100% [95% CI = 44–100%]) and accurately ruled out nystagmus in the remaining 20 traces (specificity = 100% [95% CI = 84–100%]). Positive predictive value and negative predictive value were 100% (95% CI = 44–100%) and 100% (95% CI = 84–100%), respectively. Based on these data, the accuracy of the EyePhone is estimated to be 100% (95% CI = 86–100%). A representative vertical eye position trace from a patient with posterior canal BPPV is shown in Figure 2.

Discussion

To the best of our knowledge, this pilot study is the first instance of using an ARKIT-based smartphone eye tracking to successfully record eye movements during positional testing. Moreover, we demonstrated that by

using the eye position traces obtained by an iPhone and using the MATLAB algorithm embedded in our app, we could detect positional nystagmus with reasonable accuracy during standard positional maneuvers. The app's high accuracy (100%) and predictive values can make it a promising tool for clinical and telemedicine applications, particularly in resource-limited settings in the future. As mentioned above, the EyePhone app utilizes eye position traces obtained from front camera videos of eye movement recording using Apple's (Cupertino, CA) ARKIT API [20]. While the details of how ARKIT's algorithm estimates eye position have not been made public by Apple (Cupertino, CA), previous studies by our team and other teams have examined its accuracy in detecting eye and head position, finding it to be comparable to standard reference methods [16, 20]. Furthermore, we have previously published on using the same MATLAB algorithm that we used in this study to detect and characterize the direction and velocity of induced optokinetic nystagmus. In that study, we showed a high correlation between slow phase velocities obtained from EyePhone traces and FDA-approved standard-of-care VOG goggles ($r = 0.98$ for horizontal and $r = 0.94$ for vertical) [19]. The accuracy of EyePhone in detecting positional nystagmus in the present study aligns with findings from our prior study, where we showed an area under the receiver operator curve of 0.87 for

detecting horizontal and vertical nystagmus compared to VOG [19].

Smartphone eye tracking is not limited to iPhone-based ARKIT application. Friedrich et al. [22] have previously published on using a modified convolutional neural network called ConVNG to detect slow phase velocity of optokinetic nystagmus with an accuracy on par with VOG. Similarly, Reinhardt et al. explored the use of convolutional neural networks on videos obtained by an Android-based smartphone (Samsung Galaxy S10) for detecting nystagmus during caloric testing. Similar to our study, they compared the AI-detected nystagmus videos to expert reviews showing a high sensitivity (90%) with a lower specificity (32%) in detecting caloric nystagmus [23].

Despite the small sample size of our study, the app achieved very high specificity and predictive values, further supporting its feasibility for positional nystagmus detection. Prior studies highlight challenges in diagnosing the underlying causes of vertigo, particularly in emergency settings where clinical misdiagnosis remains a concern [4, 24]. Our findings suggest that smartphone-based tools could complement bedside tests like HINTS (head impulse test, nystagmus, test of skew deviation), potentially increasing real-world diagnostic accuracy and accessibility. Furthermore, the app's potential extends to telemedicine and remote patient monitoring. Integrating automated algorithms for detecting positional nystagmus, as demonstrated in our study, could further streamline clinicians' workflow. While our study focused on positional nystagmus, future work could expand to other vestibular disorders, such as vestibular neuritis or Ménière's disease, where eye movement analysis plays a crucial role [25]. In the future, combining smartphone eye tracking with AI-based pattern recognition have the potential to enhance diagnostic capabilities across the vestibular spectrum.

Limitations

Our sample size was small, and findings may not be generalizable to a broader population. Examiners obtaining videos with the EyePhone app were familiar with the app, and performance might vary in the hands of less skilled examiners. Reliance on expert video review as the gold standard introduces potential observer bias. Larger studies are needed to confirm these preliminary results and explore accuracy and utility.

Conclusion

This pilot study demonstrated that a smartphone eye-tracking application can be used to produce eye position traces that accurately indicate positional nystagmus, offering a cost-effective, scalable alternative to specialized equipment. The app holds significant potential for improving vestibular diagnostics, particularly in resource-limited and telemedicine settings. Given the promising results of this pilot study, the next step for our team is to test the accuracy of the app in a larger population of patients and when used by multiple providers.

Statement of Ethics

This study protocol was approved by the Johns Hopkins Institutional Review Board (IRB00258938). Written informed consent was obtained from all participants.

Conflict of Interest Statement

David E. Newman-Toker and Jorge Otero Millan have an approved patent regarding the use of EyePhone in tracking eye and head position. David E. Newman-Toker, Ali Saber Tehrani, Jorge Otero Millan, Hector Rieiro, and Pouya B. Bastani have a provisional patent application regarding using the EyePhone for recording saccades and smooth pursuit. The other authors have no conflicts of interest to declare.

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Author Contributions

V.P., P.B.B., D.E.N.-T., H.R., J.O.-M., and A.S.T. contributed to the study's conception and design. V.P., P.B.B., J.O.-M., H.R., D.E.H., D.S.Z., and A.S.T. contributed to the acquisition and analysis of the data. V.P., P.B.B., D.S.Z., and A.S.T. contributed to drafting a significant portion of the manuscript.

Data Availability Statement

Due to institutional regulations, the data supporting the findings of this study are not publicly available. However, aggregate de-identified data are available upon reasonable request from the corresponding author (A.S.T.).

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